Innovation

Magnetic Resonance Imaging (MRI) is the standard modality for evaluating craniopharyngiomas (CPs) and their relationship to adjacent hypothalamic and optic structures. Precise delineation of both the tumor and surrounding anatomy enables quantitative assessments that guide diagnosis, surgical planning, and longitudinal monitoring. However, manual segmentation remains labor-intensive and subject to high inter-observer variability, limiting reproducibility and scalability in both research and clinical workflows.

Automated segmentation enables population-level imaging analyses by supporting the construction of standardized atlases and radiomic databases. This scalability increases statistical power in clinical studies and accelerates the development of evidence-based surgical guidelines, ultimately improving our understanding of hypothalamic involvement in both healthy and disease states.

Deep learning approaches—especially convolutional and transformer-based architectures—have achieved expert-level accuracy in medical image segmentation across diverse datasets. Yet, few models address the unique challenges of the sellar–suprasellar region, where small structures, heterogeneous contrast, and mass effect complicate delineation. Our project extends these advances to craniopharyngiomas, leveraging data augmentation and multi-contrast MRI to enhance model generalization despite limited sample sizes.

By linking automated imaging biomarkers to operative and endocrine outcomes, our work bridges radiology and clinical decision-making. The resulting segmentation framework will enable hypothalamus-sparing surgical planning, improve prediction of postoperative endocrine dysfunction, and establish a reproducible foundation for longitudinal outcome studies.

References

1. Calvanese, F., Matarazzo, P., Massimi, L., Di Rocco, C., & Frassanito, P. (2022). *Neoadjuvant BRAF and MEK inhibitor targeted therapy in craniopharyngiomas: Experience and review of the literature. Frontiers in Oncology, 12,* 907301. https://doi.org/10.3389/fonc.2022.907301
2. Javidialsaadi, M., Nikouei, F., Rezaei, S., & Omidvari, S. (2025). *Advances in the management of craniopharyngioma: A comprehensive review. Journal of Clinical Medicine, 14*(4), 1101. https://doi.org/10.3390/jcm14041101
3. Jiang, C., Zhang, W., Wang, H., Jiao, Y., Fang, Y., Feng, F., Feng, M., & Wang, R. (2023). *Machine learning approaches to differentiate sellar-suprasellar cystic lesions on magnetic resonance imaging. Bioengineering, 10*(11), 1295. https://doi.org/10.3390/bioengineering10111295
4. Khan, D. Z., Lindner, T., Bander, E. D., et al. (2024). *Artificial-intelligence-assisted operative anatomy recognition in sellar region surgery. NPJ Digital Medicine, 7*(1), 104. https://doi.org/10.1038/s41746-024-01273-8
5. Lohkamp, L. N., De Cloet, F., Kehler, U., & Merchant, T. E. (2023). *An update on multimodal management of pediatric craniopharyngioma: Recent advances and knowledge gains. Frontiers in Oncology, 13,* 1101652. https://doi.org/10.3389/fonc.2023.1101652

6. Qu, H., Ban, Q., Zhou, L., Duan, H., Wang, W., & Peng, A. (2025). *Radiomic study of common sellar region lesion differentiation in MRI based on a multi-classification machine-learning model. BMC Medical Imaging, 25,* 147. https://doi.org/10.1186/s12880-025-01690-5
7. Shu, X., Zhou, Y., Li, F., Zhou, T., Meng, X., Wang, F., Zhang, Z., Pu, J., & Xu, B. (2021). *Three-dimensional semantic segmentation of pituitary adenomas based on the deep learning framework nnU-Net: A clinical perspective. Micromachines, 12*(12), 1473. https://doi.org/10.3390/mi12121473
8. Wang, H., Feng, M., Zhang, Y., Jiao, Y., Wang, R., & Feng, F. (2021). *Development and evaluation of deep-learning-based automatic segmentation of the sellar region in MRI and its clinical usefulness. Frontiers in Endocrinology, 12,* 693607. https://doi.org/10.3389/fendo.2021.693607
9. Yan, X., Liu, Z., & Wang, F. (2023). *Deep-learning-based automatic segmentation and detection of craniopharyngiomas and surrounding structures in MRI. Frontiers in Oncology, 13,* 1164305. https://doi.org/10.3389/fonc.2023.1164305